

GARAGE RENOVATION

CONCEPT DRAWING SET

Architecture · structure · electrical · data · shop air

BASIS OF SET

Current structural model: frame-20260920.01-beam-scheme.compas.json
Current architectural/equipment layout: cabinets.py
Units: inches. North is up on plans unless noted.
Member and equipment locations are taken directly from the current model.
Electrical, Ethernet, compressed-air and dust routes are conceptual.

DRAWING INDEX

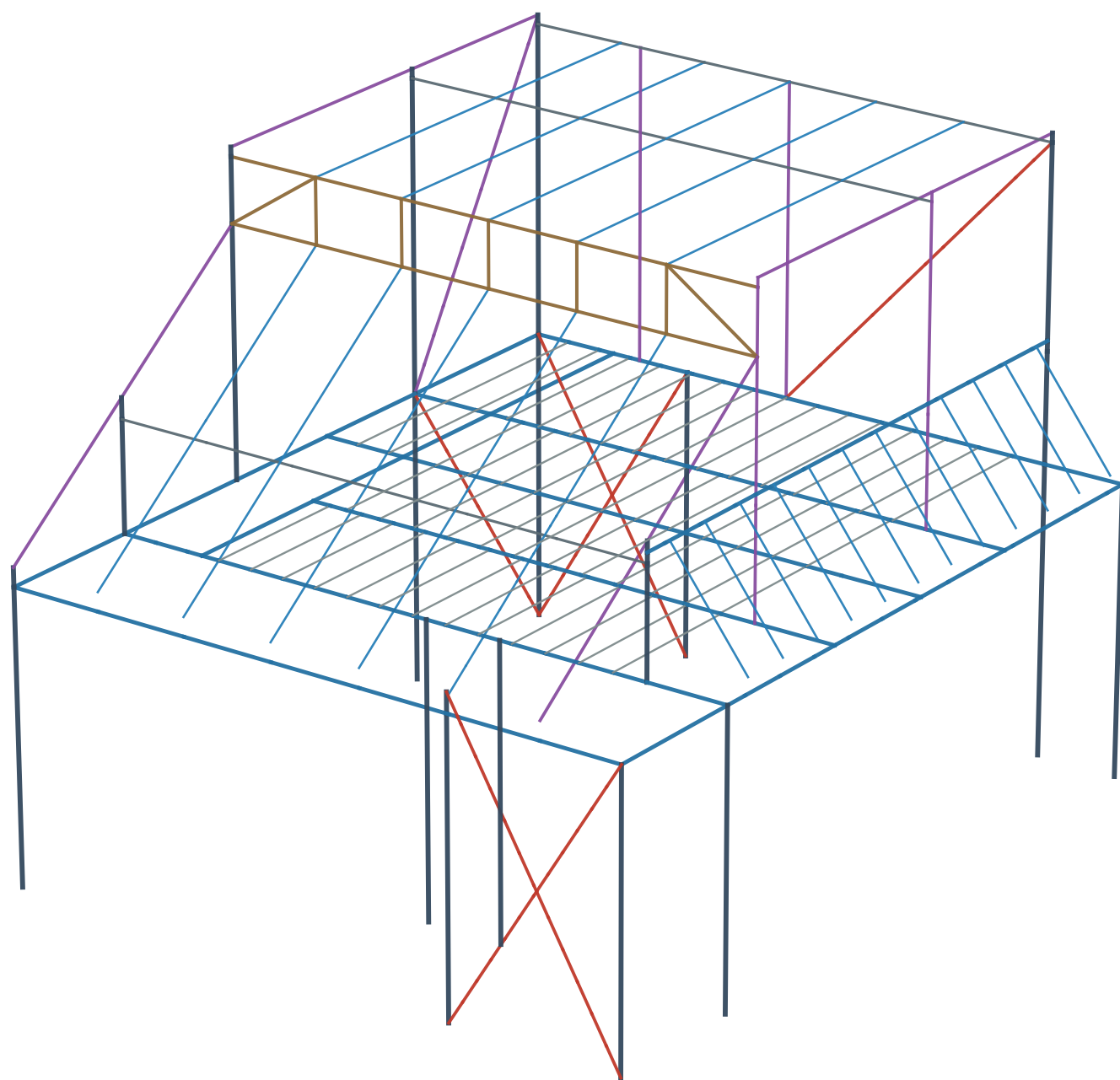
G0.01 Cover / drawing index
A1.01 Isometric and exterior elevations
A1.02 Loft equipment and cabinet plan
A1.03 Ground-floor and shed equipment plan
E1.01 Concept electrical distribution
T1.01 Concept Ethernet distribution
M1.01 Concept dust collection and compressed air

USE AND LIMITATIONS

This set records design intent for coordination and discussion. It is not a permit set, fabrication package, or construction document.
A licensed design professional must verify dimensions, loads, connections, foundations, fire/life-safety work and all MEP requirements before construction.
Proposed service routes show endpoints and zoning; conductor, cable, pipe and duct sizes remain to be designed.

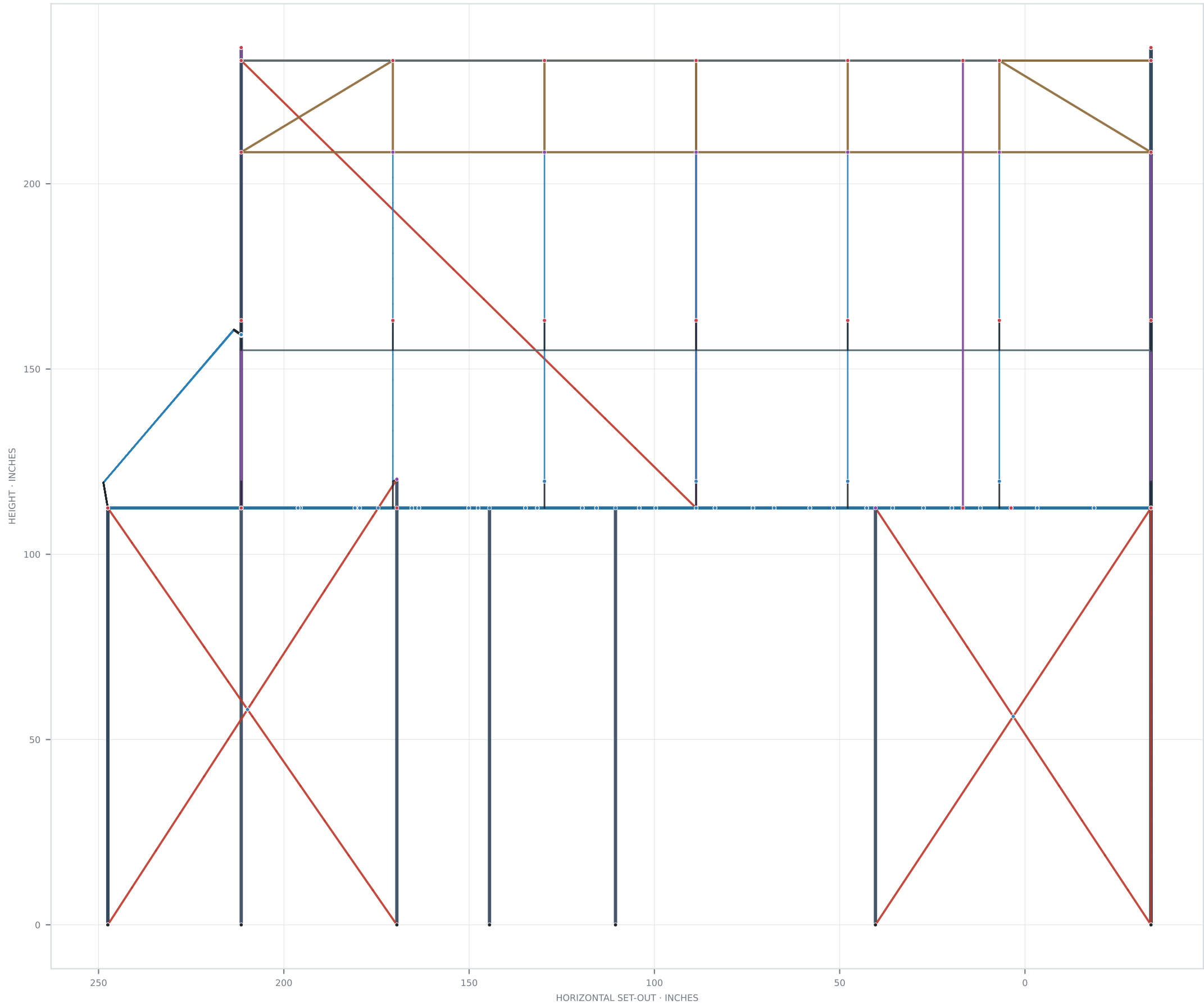
ISOMETRIC — CURRENT FRAME

SCALE: DIAGRAMMATIC



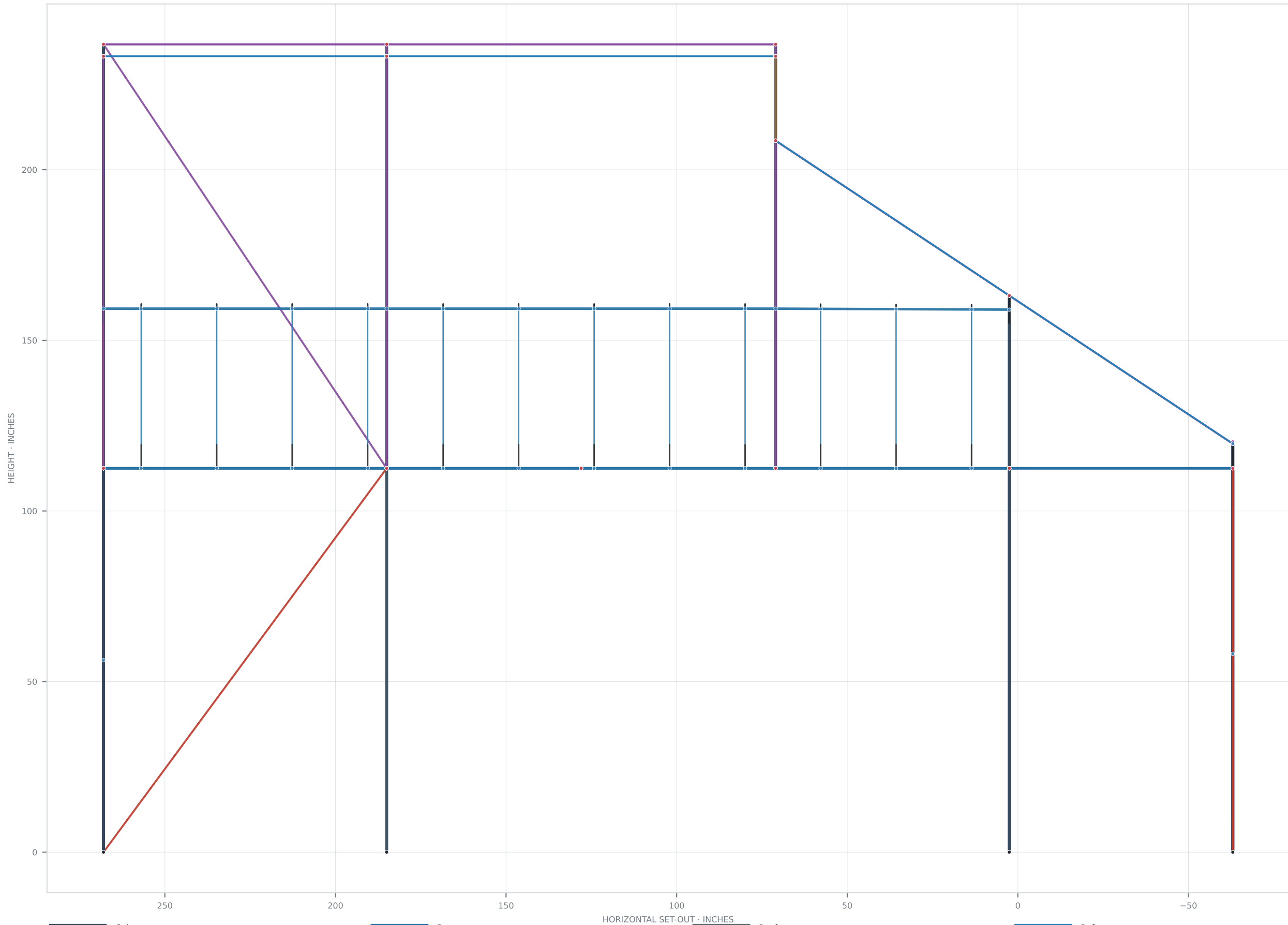
NORTH ELEVATION — LOOKING SOUTH

SCALE: DIAGRAMMATIC



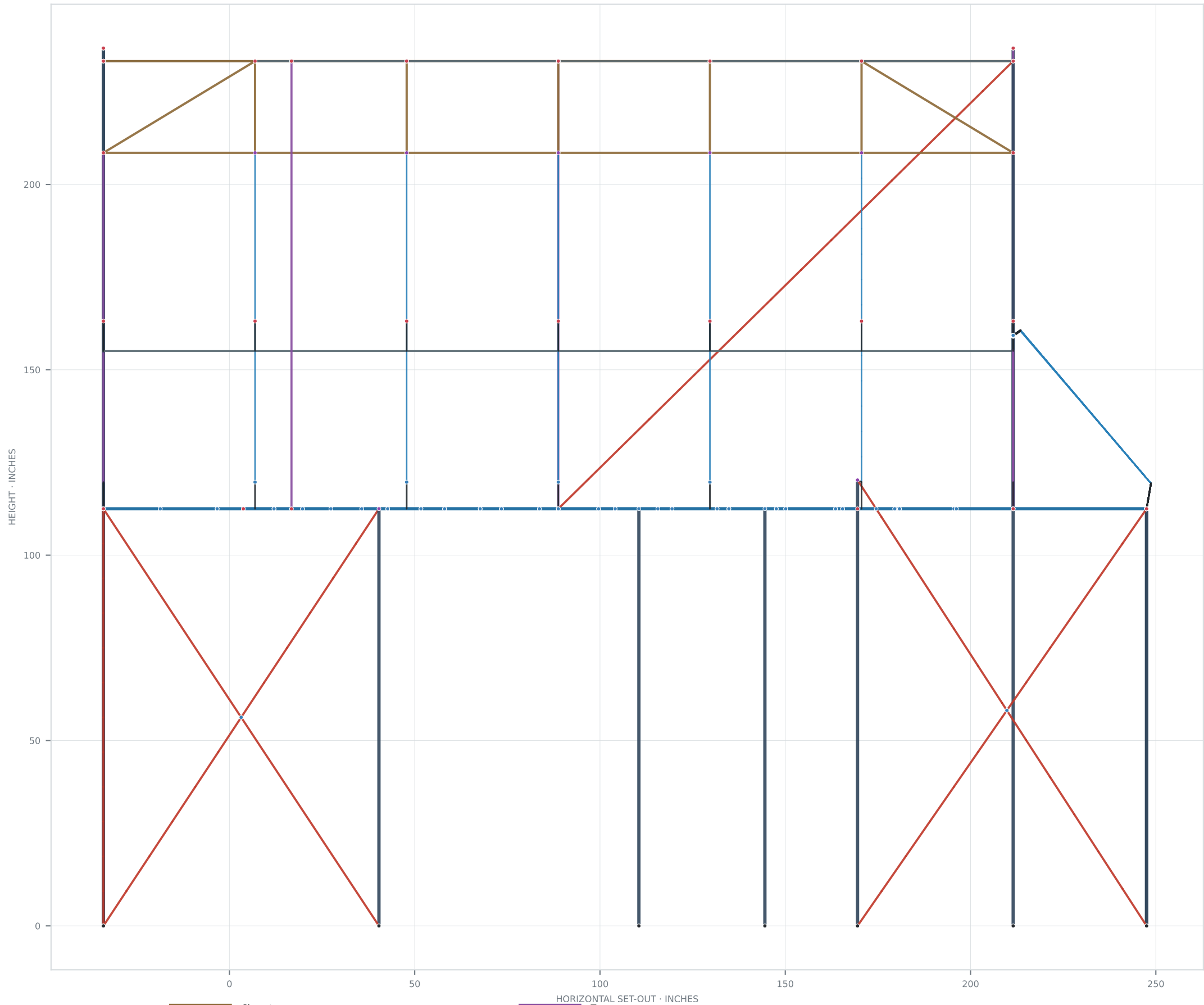
EAST ELEVATION — LOOKING WEST

SCALE: DIAGRAMMATIC



SOUTH ELEVATION — LOOKING NORTH

SCALE: DIAGRAMMATIC



COLOR KEY

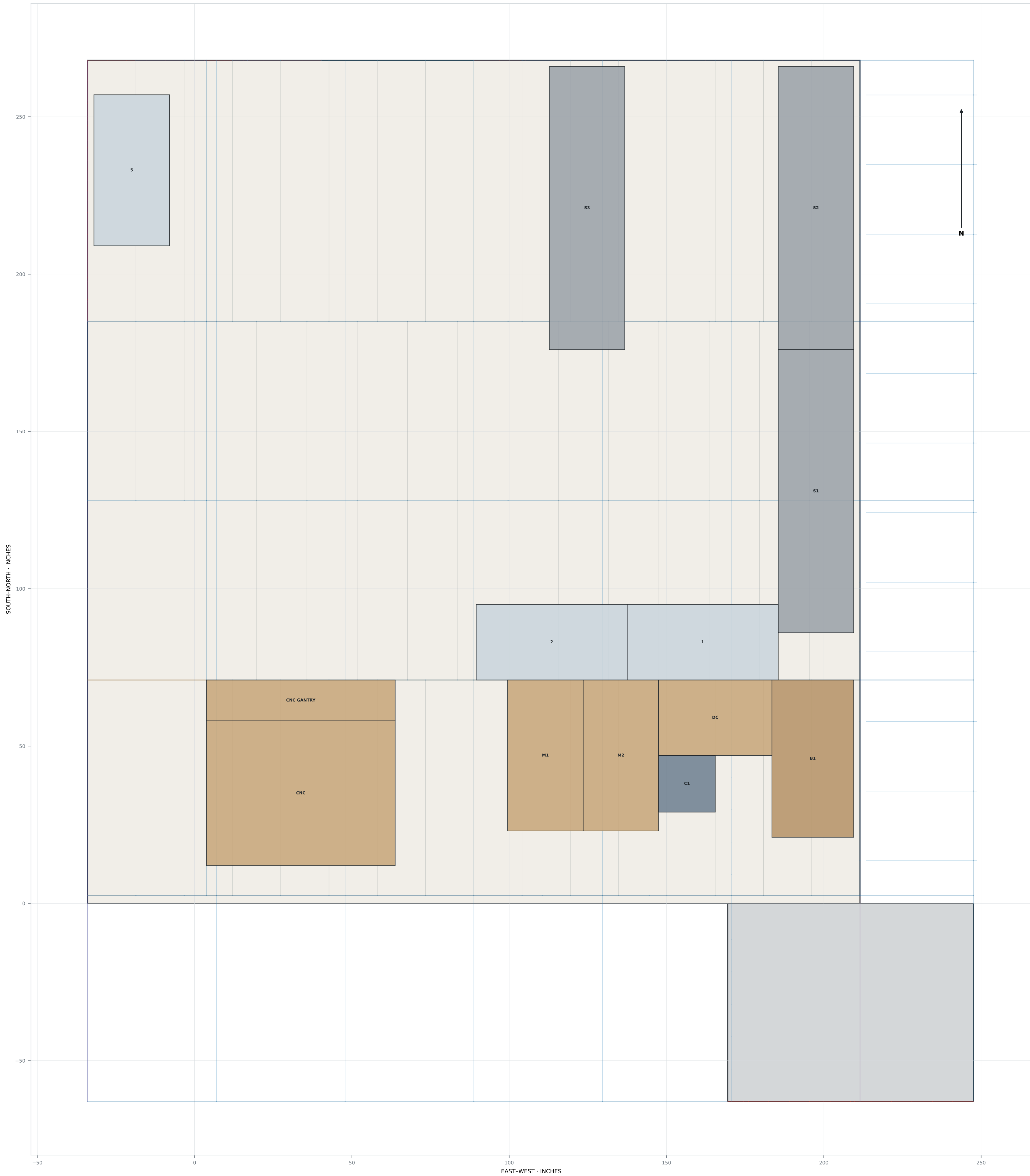
- Columns
- Beams
- Roof
- Rafters
- Joists
- Bracing
- Clerestory
- Truss

JOINTS

- Welded — moment-continuous
- Welded joint with pinned attachments
- Pinned — bolted, seated or hung
- Pinned base — anchor rods

LOFT PLAN

SCALE: DIAGRAMMATIC



LOFT EQUIPMENT

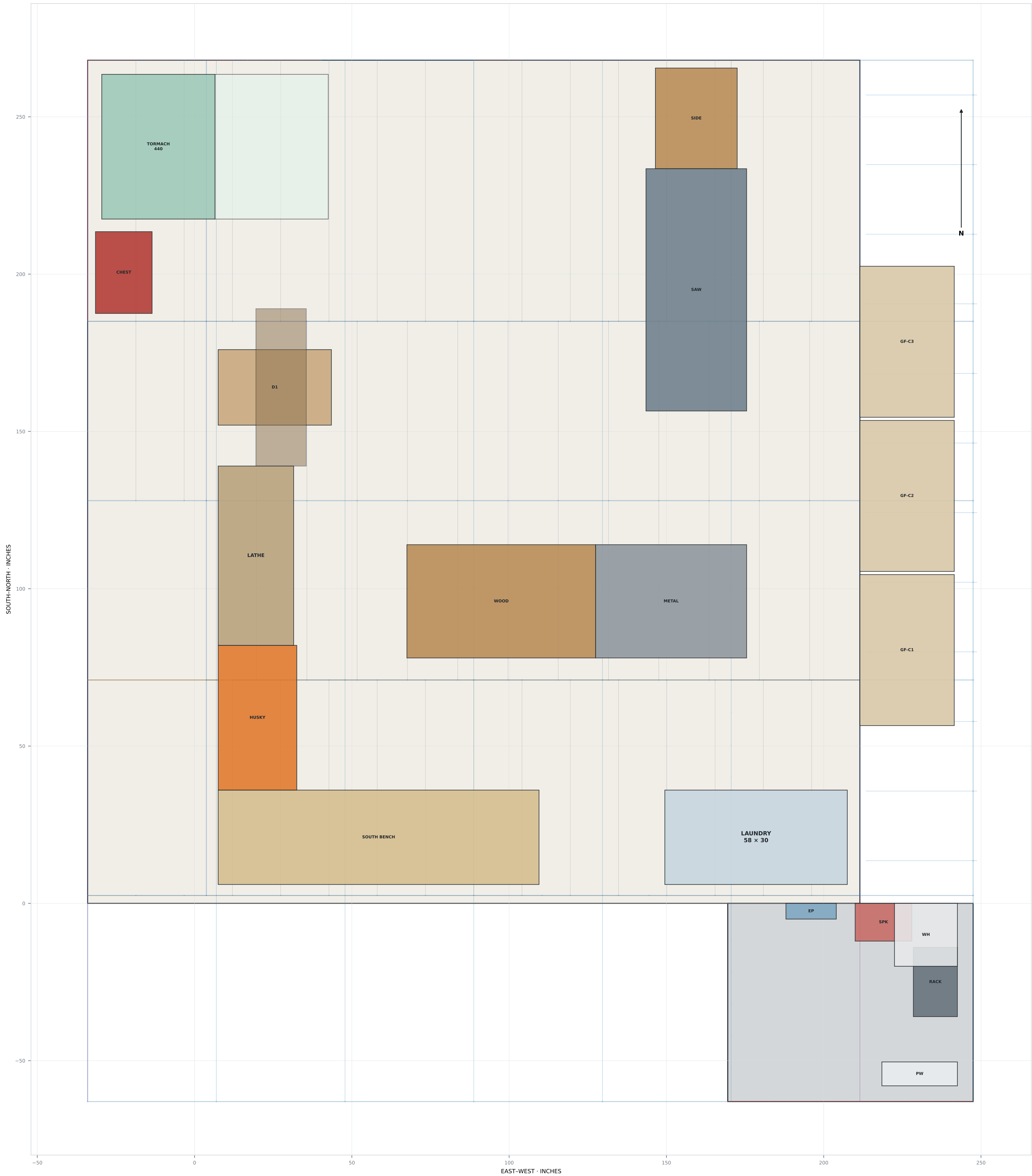
- 1 cabinet 48 × 24 × 72 in
- 2 cabinet 48 × 24 × 72 in
- 5 cabinet 24 × 48 × 64 in
- B1 crate 26 × 50 × 46 in
- M1 machine 24 × 48 × 40 in
- M2 machine 24 × 48 × 40 in
- S1 shelving 24 × 90 × 90 in
- S2 shelving 24 × 90 × 90 in
- S3 shelving 24 × 90 × 90 in
- C1 compressor 18 × 18 × 36 in
- DC machine 36 × 24 × 60 in
- CNC machine 60 × 46 × 37 in
- CNCg machine 60 × 13 × 53.25 in

COORDINATION NOTES

Cabinets and equipment are nonstructural.
Maintain the documented walkways and access zones.
Verify anchorage, working clearances and concentrated floor loads.
CNC is shown in its parked position.

GROUND-FLOOR PLAN

SCALE: DIAGRAMMATIC



GROUND-FLOOR ELEMENTS

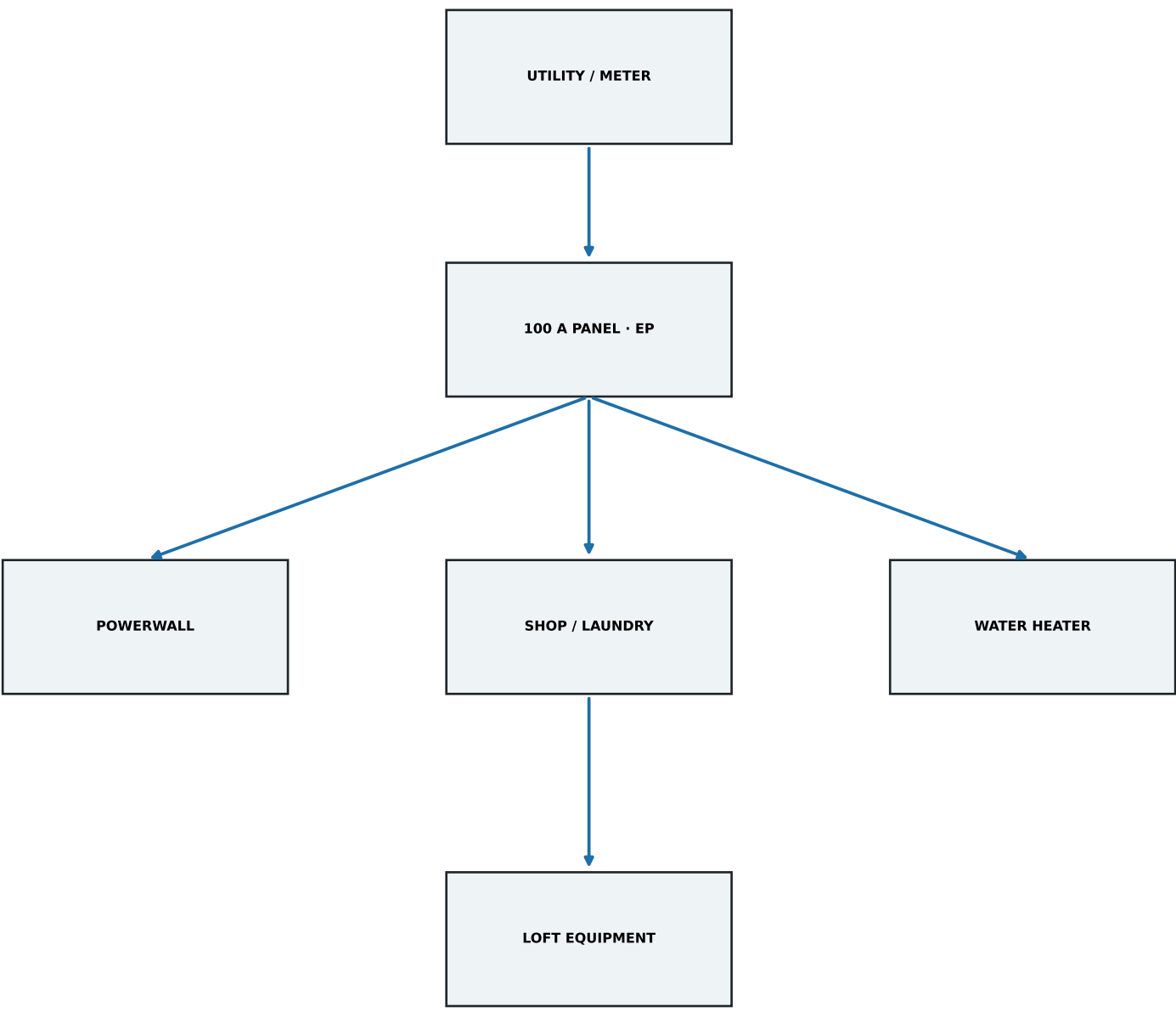
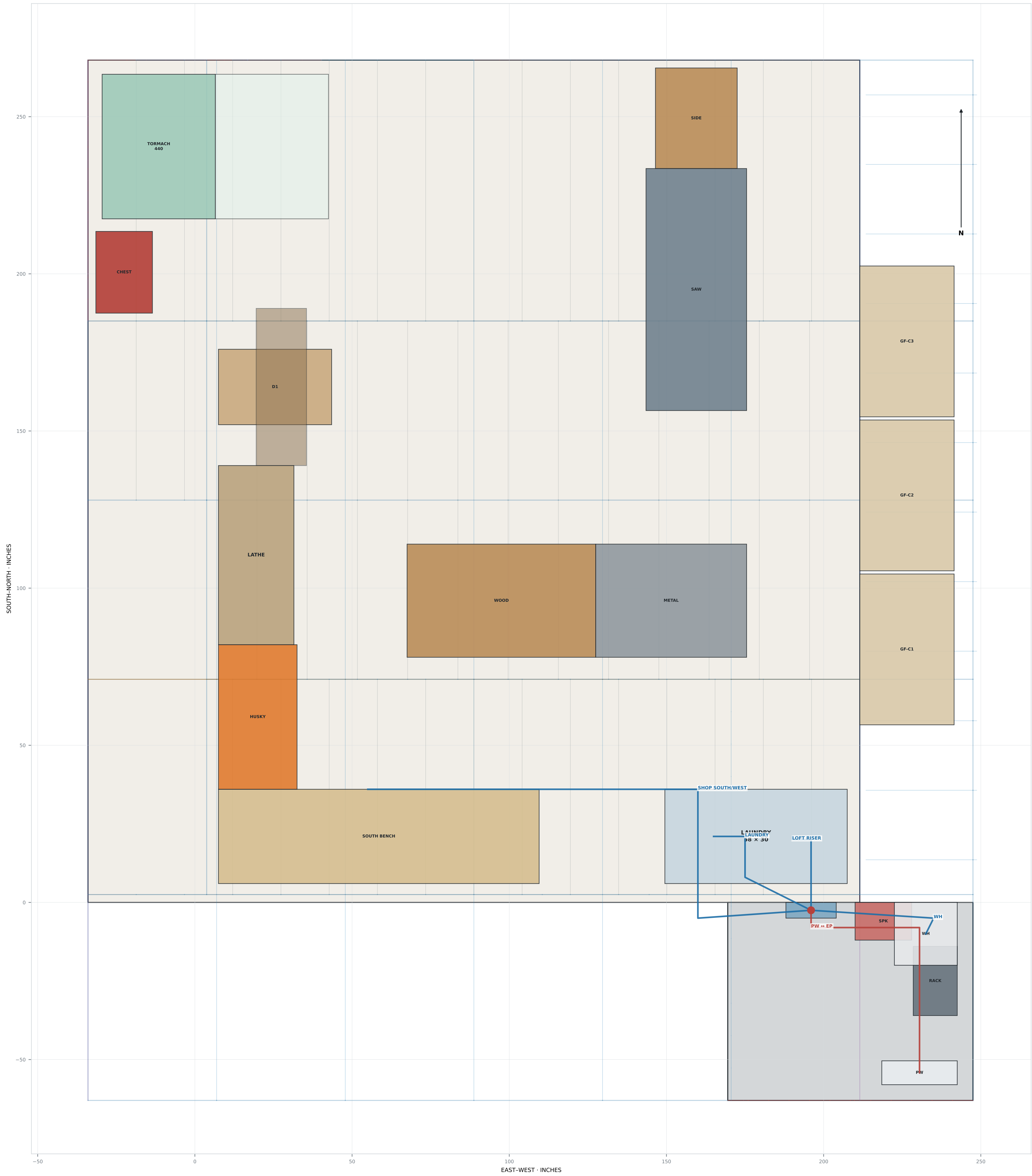
GF-C1-C3 east built-ins
LAUNDRY 58 x 30 x 66 in
SOUTH BENCH 30 in deep
CHEST Craftsman 26 x 18 x 58 in
HUSKY rolling cabinet 46 x 25 x 36 in
D1 RF-30 mill/drill + 50 in. table
WOOD / METAL tables 60 x 36, 48 x 36
SAW saw/router 32 x 77 x 38 in
SIDE table 26 x 32 in
LATHE 24 x 57 in
PW Tesla Powerwall 3 · 291 lb · moved from the loft wall
EP 100 A electrical panel · conceptual 16 x 5 x 30 in. envelope
SPK sprinkler equipment cabinet · conceptual 18 x 12 x 24 in. envelope
RACK wall-mounted 19 in. rack · conceptual 12U · 14 in. deep
WH 15 gal electric water heater · 20 in. diameter x 29 in. high

COORDINATION NOTES

Shed equipment is shown at the current cabinet-layout positions.
Keep required electrical-panel, heater and sprinkler service clearances.
Confirm wall construction and equipment anchorage before installation.

POWER PLAN — CONCEPT ROUTING

SCALE: DIAGRAMMATIC

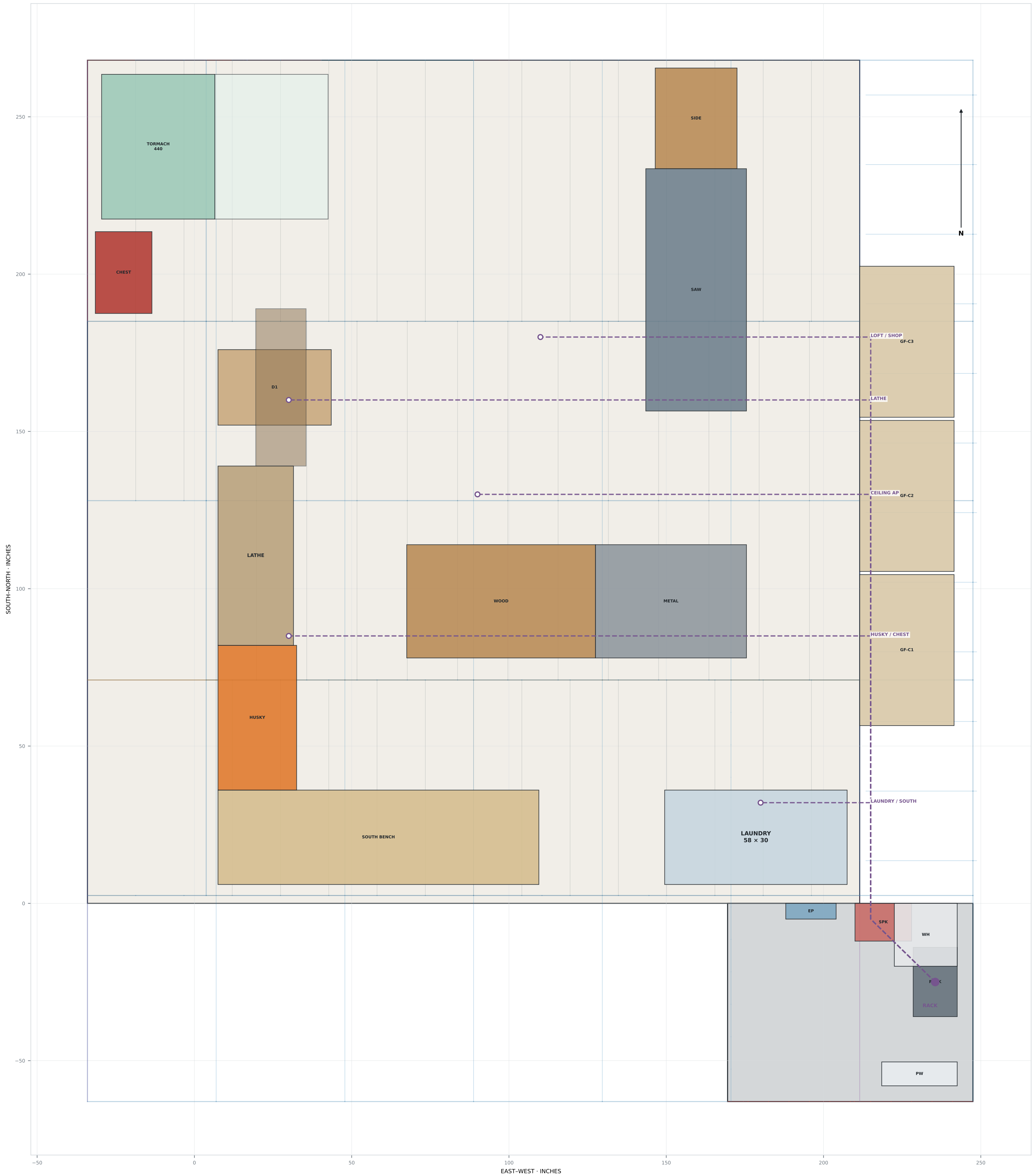


DESIGN DECISIONS REQUIRED

Confirm service point, available fault current and Powerwall topology.
Develop load calculation and panel schedule; size all feeders and branches.
Provide dedicated circuits as required for laundry, water heater, CNC, mill/drill, dust collector and compressor.
Coordinate disconnects, GFCI/AFCI protection, grounding, lighting, receptacle spacing and code working clearances.
All routes shown are conceptual home-run zones.

DATA PLAN — CONCEPT HOME RUNS

SCALE: DIAGRAMMATIC



SYSTEM CONCEPT

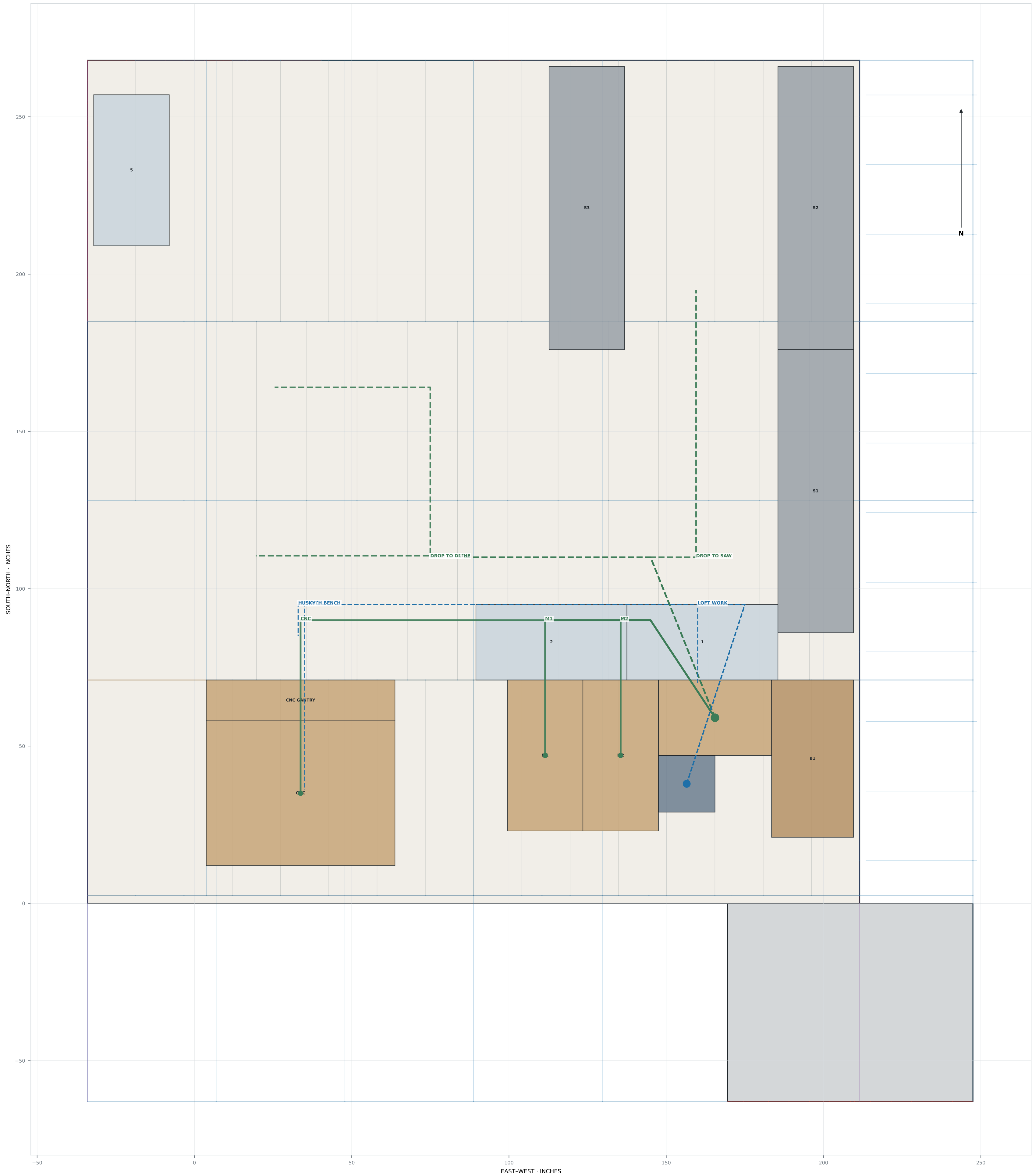
The wall-mounted 19-inch rack in the shed is the star point. Provide home-run cabling to shop work zones, the lathe area, loft equipment and a ceiling wireless-access-point location. Use pathway and spare capacity for future cameras, controls and access points.

COORDINATION NOTES

Routes are schematic; locate pathways after power and fire-protection coordination. Maintain separation from line-voltage conductors and sources of interference. Confirm cable category, patch-panel size, PoE load, rack ventilation, bonding and surge protection. Label both ends and record final cable IDs on an as-built schedule.

LOFT / SHOP SERVICES — CONCEPT ROUTING

SCALE: DIAGRAMMATIC



DUST COLLECTION CONCEPT

DC is the loft collector. Green lines show a conceptual main with branches to M1, M2 and CNC plus drops to the ground-floor lathe, D1 and saw. Provide a blast gate at every branch and a cleanout at low points. Final duct diameters, fan static pressure, filtration, grounding/bonding and fire protection require design from actual machine data.

COMPRESSED-AIR CONCEPT

C1 is the loft compressor. Blue dashed lines show a loop/drop concept serving the loft and both principal ground-floor benches. Slope mains to drains; provide isolation, filtration/regulation and flexible machine connections. Verify compressor ventilation, vibration isolation, relief discharge and receiver access.